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Longitudinal Autoantibody and Proteomic Profiles Uncover Biomarkers and Mechanisms of Disease Progression in Systemic Sclerosis

Running Head: Autoantibody and Proteomic Profiles in SSc Progression

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ABSTRACT

Objective. To evaluate dynamic changes in autoantibody and proteomic profiles in treatment-naïve systemic sclerosis (SSc) patients and identify biomarkers and mechanisms associated with disease progression.

Methods. Serum samples from 30 baseline and 49 follow-up SSc patients, along with 38 controls, were analyzed. Autoantibody profiles were assessed using an autoantigen microarray targeting 120 autoantibodies, while proteomic analysis was conducted via liquid chromatography–mass spectrometry in data-independent acquisition mode. Cox proportional hazards models were used to assess associations with disease progression, and Spearman's correlation was applied to analyze antibody-protein interactions.

Results. Baseline autoantibody profiling identified two patient subgroups: high-expression (SSc_high) and low-expression (SSc_low), though their clinical characteristics were similar. Longitudinal analysis revealed that 88.9% of patients with rising autoantibody titers experienced disease progression, while 60% of those with decreasing titers showed clinical improvement. Notably, increased CENP-B titers (HR = 5.036) were strongly linked to disease progression. Proteomic analysis identified 112 proteins involved in tissue repair and immune modulation in patients with declining titers and clinical improvement, whereas 46 proteins linked to lipid metabolism and oxidative stress were enriched in those with rising titers and disease progression. SERPINA10 emerged as the strongest risk factor for progression, while ENTPD5 was most protective. Antibody-protein correlations highlighted key molecular interactions, including SERPINE1 and muscarinic receptor antibodies, were identified.

Conclusion. Dynamic changes in autoantibody titers, rather than baseline levels, were more strongly associated with SSc trajectory. Proteomic signatures suggest lipid metabolism and oxidative stress as potential drivers, highlighting novel biomarkers and therapeutic targets for personalized SSc management.

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INTRODUCTION

Systemic sclerosis (SSc) is a rare, clinically heterogeneous autoimmune disease characterized by inflammation, progressive fibrosis of the skin and internal organs, and microvascular damage. These pathological features can lead to life-threatening complications affecting the lungs, heart, gastrointestinal tract, and kidneys. The unpredictable nature of the disease presents significant challenges for effective management, highlighting the need for reliable biomarkers to improve patient outcomes.

Previous studies have underscored the critical role of autoantibodies in predicting SSc prognosis. Key examples include anti-centromere (ACA), anti-topoisomerase I (Scl-70/ATA), and anti-RNA polymerase III (ARA) antibodies. Patients with Scl-70 antibodies face a higher risk of severe complications, such as diffuse cutaneous sclerosis and pulmonary fibrosis, whereas those with ARA have a tenfold increased risk of scleroderma renal crisis[1]. Additionally, anti-PRMT5 has recently been proposed as a potential biomarker of disease trajectory, while others—such as anti-TRIM21, anti-IFI16, anti-gephyrin, anti-gAChR, and anti-SSSCA1—have been linked to pulmonary arterial hypertension, interstitial lung disease (ILD), gastrointestinal dysmotility, cancer risk, and cardiac manifestations[2]. However, up to 11% of SSc patients test negative for antinuclear antibodies (ANA)[3], and as many as 17% lack detectable levels of established SSc-specific antibodies[4], underscoring the need for additional biomarkers to improve disease classification and prognosis. Emerging research has identified autoantibodies with potential pathogenic roles, including anti-platelet-derived growth factor receptor antibodies, anti-endothelial cell antibodies, and anti-fibroblast antibodies. Some autoantibodies also influence the transcriptomic landscape of SSc blood cells. For instance, patients with Scl-70 antibodies exhibit a pro-inflammatory and fibrotic transcriptional signature characterized by impaired tissue remodeling and elevated carnitine metabolism, whereas ACA-positive patients display a signature associated with immunomodulation, tissue homeostasis, and increased phospholipid metabolism[5]. Single-cell RNA sequencing has revealed distinct gene expression profiles in specific cell clusters across autoantibody-defined SSc subgroups[6], suggesting the potential of autoantibody profiling as a prognostic biomarker. However, further studies are required to elucidate the mechanistic role of autoantibodies in the pathogenesis of

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SSc. Exploring the pathogenic roles of autoantibodies may provide valuable insights for identifying new therapeutic targets.

Moreover, most studies on autoantibodies in SSc have been cross-sectional, leaving the relationship between autoantibody dynamics and disease progression poorly understood.

In this study, we conducted a comprehensive analysis of autoantibody levels and proteomic profiles in 30 newly diagnosed SSc patients, who were followed for 6 to 24 months, to assess dynamic changes during treatment and identify biomarkers associated with disease progression. By integrating autoantibody and protein signatures, we aim to advance biomarker-driven strategies for managing the clinical heterogeneity of SSc and to deepen our understanding of its underlying disease mechanisms.

MATERIALS AND METHODS

Patient Recruitment

Serum samples were collected at Xiangya Hospital from 30 treatment-naïve patients with SSc at the time of diagnosis (SSc baseline, n=30), as well as from 38 age- and sex-matched healthy controls (NC). In addition, a total of 49 follow-up serum samples were collected from this same cohort at various time points after treatment initiation, including at 6 months (n = 20), 12 months (n = 16), 18 months (n = 5), and 24 months (n = 8) (Supplementary Figure 1). All patients met the 2013 ACR/EULAR classification criteria for SSc, and individuals with malignancies, infections, other autoimmune diseases, or metabolic disorders were excluded.

Disease progression was defined by any of the following criteria: (1) worsening of dermal fibrosis (≥5-point and 20% increase in the mRSS)[7]; (2) radiological deterioration of lung fibrosis, based on criteria adapted from the 2022 international guideline for progressive pulmonary fibrosis. Serial high-resolution computed tomography (HRCT) scans were interpreted by experienced thoracic radiologists, and progression was defined by one or more of the following: increased extent or severity of traction bronchiectasis or bronchiolectasis; new ground-glass opacities with associated traction bronchiectasis; new areas of fine reticulation; increased extent or coarseness of reticular abnormalities; new or increased honeycombing; or increased lobar volume loss; (3) an increase in systolic pulmonary artery pressure (sPAP) to >45 mmHg on echocardiography as a surrogate for pulmonary arterial hypertension; or (4)

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new onset of digital ulcers, renal crisis, or gastrointestinal involvement[8]. SSc improvement was defined as a ≥5-point and 20% decrease in mRSS over 12 months or radiological improvement of ILD[9].

Demographic and clinical data were collected for all participants (Supplementary Table 1-3). The study was approved by the Ethics Committee of Xiangya Hospital, Central South University (Changsha, Hunan, China; ID: 201303293), and written informed consent was obtained from all participants.

Autoantibody Profiling

Autoantibody profiling was conducted using the Systemic Autoimmune-Associated Antigen Array (PA001-V3) (Genecopoeia Inc., Rockville, MD, USA), which detects 120 autoantibodies, including controls.

On the day of hybridization, autoantigen slides were equilibrated to room temperature for 30 minutes. A 16-section frame was applied to separate individual arrays, followed by blocking with 50 μL of blocking buffer under gentle agitation for 30 minutes. Serum samples were diluted 1:100 in blocking buffer, an optimized dilution based on preliminary experiments to maximize the signal-to-noise ratio. Then, 50 μL of the diluted serum was applied to each array and incubated for 60 minutes with agitation. After incubation, slides were washed three times with 100 μL of washing buffer (PBS with 0.05% Tween 20, pH 7.4) for 5 minutes each under agitation. Cy3-labeled anti-human IgG (Jackson ImmunoResearch, West Grove, PA, USA) was applied at a 1:500 dilution, followed by a 60-minute incubation at room temperature. After further washing, the frame was removed, and the slides were briefly immersed in PBS before being spun dry. Fluorescent signals were detected using a GenePix 4000B scanner with a 532 nm laser (for Cy3). Data processing was based on net signal intensity, calculated as the average fluorescence of each antigen after subtracting local background and negative control signals. Results were normalized to internal Ig controls[10].

Proteomics Analysis

Fasting morning blood samples were collected from NC and SSc patients. Serum was separated by centrifugation at 3,000 rpm for 10 minutes at 4°C and stored at -80°C until analysis. To enhance the detection of low-abundance proteins, high-abundance proteins such as albumin and immunoglobulins were depleted using the Pierce

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Albumin Depletion Kit (Thermo Fisher Scientific). The depleted serum was then denatured with 8 M urea, alkylated with 50 mM iodoacetamide, and digested overnight with trypsin at 37°C (Thermo Fisher Scientific). Peptides were purified using Pierce C18 Spin Columns (Thermo Fisher Scientific) following the manufacturer's protocol. Liquid chromatography–mass spectrometry (LC-MS/MS) was performed in data-independent acquisition (DIA) mode using a NanoAcquity UHPLC system coupled to an Orbitrap Exploris 240 mass spectrometer (Thermo Fisher Scientific). Briefly, 0.5 μg of peptides were enriched in a trap column at a flow rate of 10 μL/min for 4 minutes, then separated on an analytical column (2 μm, Acclaim PepMap 100 C18 HPLC, 100 Å, 250 × 0.075 mm) at a constant flow rate of 300 nL/min over 150 minutes. Protein expression was quantified using DIA-NN (v1.8.1) with the UniProt human protein database (www.uniprot.org/, accessed Oct. 20, 2022) as the reference library and a precursor false discovery rate (FDR) threshold of <0.05. Data were normalized using the DEP package in R (v1.16.0).

Statistical Analysis

All statistical analyses were performed using R software. The Shapiro-Wilk test was used to assess data normality. For normally distributed data with homogeneous variances, an unpaired t-test was applied, while for non-normally distributed data or those with heterogeneous variances, the Mann-Whitney U test was used. Multiple group comparisons were conducted using the Kruskal-Wallis test, and categorical variables were analyzed using the chi-squared test or fisher's exact test. To assess whether autoantibodies and proteins were significant independent predictors of SSc disease progression, hazard ratio (HR) was calculated using the Cox proportional hazards model, implemented via the survival package. Correlations between autoantibody and protein levels were analyzed using Spearman's rank correlation coefficient. A two-sided P-value < 0.05 was considered statistically significant.

RESULTS

Identification of high- and low-expression antibody groups in SSc patients

Using an autoantigen microarray, we profiled 85 autoantibodies in serum samples from 30 treatment-naïve SSc patients (SSc baseline) and 38 NC after quality control, categorizing them into ten groups: antinuclear antibodies (e.g., Scl-70/Topo I, CENP-A, CENP-B), blood factors (Factor B/D/H/I/P, Fibrinogen IV), complement proteins

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(C1q, C3/C3a), cytokines (GDF-15, TGF-β1), cytoskeletal/extracellular matrix proteins (e.g., α-actinin, β-actin, Collagen I–VI, Vitronectin), inflammatory factors (IFN-α, TNF-α), mitochondrial-related antibodies (M2, FARS2), organ-specific antibodies (e.g., Neuropilin-1, Myelin basic protein, TPO, TTG), and other antibodies (e.g., S100B, Amyloid, Hemocyanin, AQP4)(Figure 1A). Clustering analysis identified two distinct patient subgroups: a high-expression antibody group (SSc_high) and a low-expression antibody group (SSc_low) (Figure 1A). The anti-Scl-70/Topo I antibody was detected in most SSc patients but was absent in NC. However, comparison between SSc_high and SSc_low revealed no significant differences in demographics (age, sex) or clinical characteristics, including SSc subtype, disease duration, RP, ILD, mRSS, or fibrotic laboratory markers (KL-6, ferritin) (Figure 1B-C).

Serum antibody level changes correlate with disease progression in SSc patients

To investigate changes in serum antibody levels during follow-up and their association with disease progression in SSc, we analyzed antibody dynamics in two groups: SSc_high (n = 12) and SSc_low (n = 18), classified based on average antibody density. Antibody changes were categorized into three groups based on a >20% deviation from baseline at the first follow-up: increased, decreased, or unchanged (Supplementary Table 4). Among SSc_high patients, 10 (83.33%) exhibited decreased antibody levels, 1 (8.33%) showed an increase, and 1 remained unchanged. In the SSc_low group, 8 (47.05%) patients had increased antibody levels, 5 (29.41%) showed a decrease, and 4 remained unchanged (Figure 2A-D). These findings suggest that patients with high baseline antibody levels were more likely to experience a decline over time, whereas those with low baseline levels were more prone to an increase.

To further assess the clinical relevance of antibody changes, we classified all patients into three groups based on antibody dynamics—increase (n=9), decrease (n=15), or unchanged (n=5)—and evaluated their disease progression. Among patients in the antibody decrease group, 60% showed clinical improvement, while 26.7% experienced disease progression. In contrast, 88.9% of patients in the antibody increase group exhibited disease progression, with no cases of improvement observed in either the increase or unchanged group. The difference in improved patients between the antibody increase and antibody decrease group was statistically

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significant ($p < 0.05$) (Figure 2E, Supplementary Figure 2), indicating that a decline in antibody levels was associated with clinical improvement.

To further explore the role of specific antibody changes in disease progression, we calculated HR. An increase in 26 antibodies during treatment was significantly associated with disease progression, with CENP-B showing the highest risk (HR = 5.036). Other implicated antibodies included LC1, muscarinic receptor, SP100, intrinsic factor, myosin, MPO, Factor I, GDF-15, fibrinogen S, β-actin, and CENP-A, among others (Figure 2F).

Proteomic changes associated with differential antibody dynamics and disease trajectories

To investigate the influence of antibody titers on disease progression in SSc, we analyzed serum proteomic changes in two patient cohorts: one with significantly decreased antibody titers and clinical improvement, and another with significantly increased titers and disease progression. In patients with reduced titers and improved clinical conditions, we identified 112 proteins (54 increased, 58 decreased) that exhibited significant changes, whereas these proteins remained unchanged in patients with elevated titers and disease progression. Functional analysis revealed that these proteins were primarily involved in leukocyte migration (CD74, ICAM1), differentiation (IHH, LGALS1), chemotaxis (PLA2G7, SERPINE1), extracellular matrix organization (VIM, NID2), collagen biosynthesis and metabolism (TGFBI, IHH), and endothelial cell proliferation (LRG1, PDGFB, SPARC) (Figure 3A-B, Supplementary Table 5-6). Conversely, in patients with increased titers and disease progression, we identified 46 distinct protein changes (16 increased, 30 decreased) that were absent in the improved group; these proteins were predominantly associated with lipid metabolism (ADIPOQ, ANXA2, APOA4, APOE), including the regulation of sterol and cholesterol transport, lipid localization, triglyceride metabolism, and cholesterol efflux, as well as with acute inflammatory responses (SAA1, SAA2, SAA4) and cellular responses to oxygen radicals and superoxide (SOD2). Additionally, seven proteins (1 increased, 6 decreased) common to both groups were linked to glycolytic processes and pyridine/purine nucleotide catabolism (LDHA) (Figure 3A-B, Supplementary Table 5-6).

HR analysis identified SERPINA10, SERPINE1, C1QB, C8G, and MINPP1 as risk factors for disease progression, whereas ENTPD5, HYOU1, LRG1, MMRN2, and

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CLU were associated with protection against progression (Figure 3C). Additionally, correlation analysis revealed that antibody changes were positively correlated with risk proteins and negatively correlated with protective proteins. Figure 3D highlights the top antibody-protein pairs, with SERPINE1, LDHA, FAH, APOA2, and BID as the proteins most frequently associated with antibody correlations, while muscarinic receptor, collagen VI, and aggrecan were the antibodies most frequently linked to protein correlations.

DISCUSSION

In this study, we identified two distinct groups of patients with SSc based on their baseline autoantibody expression levels. However, these baseline levels did not significantly correlate with clinical characteristics, such as the mRSS, ILD, or RP, suggesting that static antibody levels at diagnosis may not accurately reflect disease severity. Instead, our longitudinal analysis demonstrated that dynamic changes in antibody titers were more predictive of disease progression. Patients with rising antibody titers were more likely to experience disease worsening, whereas those with declining titers showed clinical improvement. We hypothesize that rising antibody titers during treatment may reflect ongoing immune activation, insufficient therapeutic response, or impending flare, whereas decreasing titers may signal immune modulation and treatment efficacy. Additionally, antibody kinetics and clinical manifestations may be temporally misaligned. High antibody titers or active immune response at baseline may not always translate to severe fibrotic outcomes, particularly if patients receive timely and effective treatment that mitigates downstream tissue damage.

To gain deeper mechanistic insights, we integrated autoantibody data with proteomic analysis. Patients with decreasing autoantibody titers and clinical improvement exhibited proteomic signatures associated with tissue repair and immune modulation. In contrast, those with increased titers and disease progression showed enrichment in proteins linked to lipid metabolism and oxidative stress. These findings suggest that persistent dysregulation of lipid metabolism and oxidative stress may drive disease progression. Targeting these pathways—such as correcting lipid metabolism abnormalities—could offer therapeutic benefits. Additionally, both progression and improvement groups exhibited alterations in glycolysis and nucleotide catabolism, indicating common molecular responses to treatment.

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Several antibodies and proteins were significantly associated with disease progression. Among them, plasma serpin family proteins—SERPINE1, SERPINA5, and SERPINA10—were particularly notable. This aligns with prior research demonstrating that elevated SERPINE1 expression in SSc skin is linked to progressive skin fibrosis[11]. Conversely, protective factors such as apolipoproteins (APOA4, APOA2), PCSK9, and oxidative stress-related proteins (SELENOP, HYOU1) were associated with reduced disease severity.

Among the autoantibodies examined, CENP-B and CENP-A demonstrated predictive value for disease progression. This finding is consistent with previous studies showing that the presence of CENP-A and/or CENP-B predicts the extent of skin involvement over time[12]. Similarly, our research highlighted the role of anti-muscarinic receptor antibodies in SSc progression, with strong correlations observed between these antibodies and multiple serum proteins. Earlier studies have linked gastrointestinal dysmotility in SSc to circulating autoantibodies against the muscarinic-3 receptor, which inhibit smooth muscle contraction when stimulated by cholinergic agents and block indirect muscle responses triggered by electric field neural stimulation[13]. These results suggest that anti-muscarinic receptor antibodies may contribute to SSc-related gastrointestinal dysmotility by disrupting muscarinic neurotransmission[14].

Anti-collagen antibodies reactive to types I, II, III, IV, and V have been well-documented in SSc[15]. In our study, we detected antibodies against collagen types I–VI in SSc serum and identified changes in anti-collagen VI antibodies as a risk factor for disease progression. Moreover, anti-collagen VI antibodies showed strong correlations with several serum proteins, including SERPINE1, LDHA, and APOA4, among others. However, their precise relationships and potential roles in SSc pathogenesis require further investigation in future studies.

This study has several limitations. First, the relatively small sample size and the decline in patient numbers over time, particularly at the 24-month follow-up, limit the statistical power and the strength of longitudinal conclusions. Additionally, the predominance of ATA-positive patients in this cohort results in a relatively homogeneous autoantibody distribution, which may reduce the generalizability of our findings to the broader SSc population. Larger, multicenter studies with more diverse autoantibody profiles are needed to validate these results and improve their applicability. Second, although we observed significant associations between changes in antibody titers, proteomic profiles, and disease progression, the causal relationships

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remain unclear. Future mechanistic studies are essential to determine how specific antibodies contribute to SSc pathogenesis at the molecular level.

In summary, our findings establish a connection between increasing autoantibody titers, proteomic changes, and the progression of SSc. By combining autoantibody and proteomic data, we deepen our understanding of SSc pathogenesis, setting the stage for biomarker-driven approaches to manage the disease. Future research should focus on validating these biomarkers in diverse populations, clarifying the roles of the identified proteins, and creating tools for real-time clinical monitoring.

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CONFLICT of INTEREST

The authors have declared no conflicts of interest.

AUTHOR CONTRIBUTIONS

All authors contributed to at least one of the following aspects of manuscript preparation: conceptualization, sample collection, investigation, methodology, formal analysis, software visualization, and/or writing. As the corresponding author, Honglin Zhu confirms that all authors have reviewed and approved the final version for publication.

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Figure 1. Antibody Expression Analysis in Baseline SSc Patients Compared to NC.

(A) Heatmap illustrating the distribution of autoantibodies in baseline SSc patients and NC.

(B) Boxplot comparing clinical characteristics between SSc_high and SSc_low.

(C) Boxplot comparing the demographics or clinical characteristics between SSc_high and SSc_low.

Figure 2. Changes in Antibody Expression During Follow-Up in SSc Patients.

(A) Heatmap depicting changes in autoantibody expression in the SSc_high group.

(B) Heatmap showing changes in autoantibody expression in the SSc_low group.

(C) Line graph illustrating changes in autoantibody levels in the SSc_high group.

(D) Line graph depicting changes in autoantibody levels in the SSc_low group.

(E) Bar chart comparing disease progression, improvement, or stability in SSc patients.

(F) Forest plot illustrating the HR of autoantibody level changes on SSc disease progression.

Figure 3. Impact of Antibody Titer Changes on Serum Proteomic Profiles.

(A) Heatmap showing changes in serum proteomic profiles in SSc patients before and after treatment.

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(B) Bubble plot representing GO pathway enrichment for altered proteins in (A).
(C) Forest plot showing the HR of altered protein levels on SSc disease progression.
(D) Bubble plot illustrating the correlation between changes in protein levels and antibody titer changes.

Supporting Information

Supplementary Figure S1

Supplementary Figure 1. Patient Enrollment, Follow-Up Schedule, and Sample Collection Timeline. A total of 30 treatment-naïve patients with SSc were enrolled at baseline. Serum samples were collected from all patients at diagnosis and during follow-up after initiation of immunosuppressive therapy.
Supplementary Figure 2. Proportion of progressors and improvers across SSc subgroups.

Supplementary Figure S2

Supplementary Table S1

Supplementary Table 2. Therapeutic approaches of SSc_low and SSc_high group
Supplementary Table 3. Therapeutic approaches of increase, decrease and unchanged group

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Supplementary Table S4

Supplementary Table S5

Supplementary Table S6

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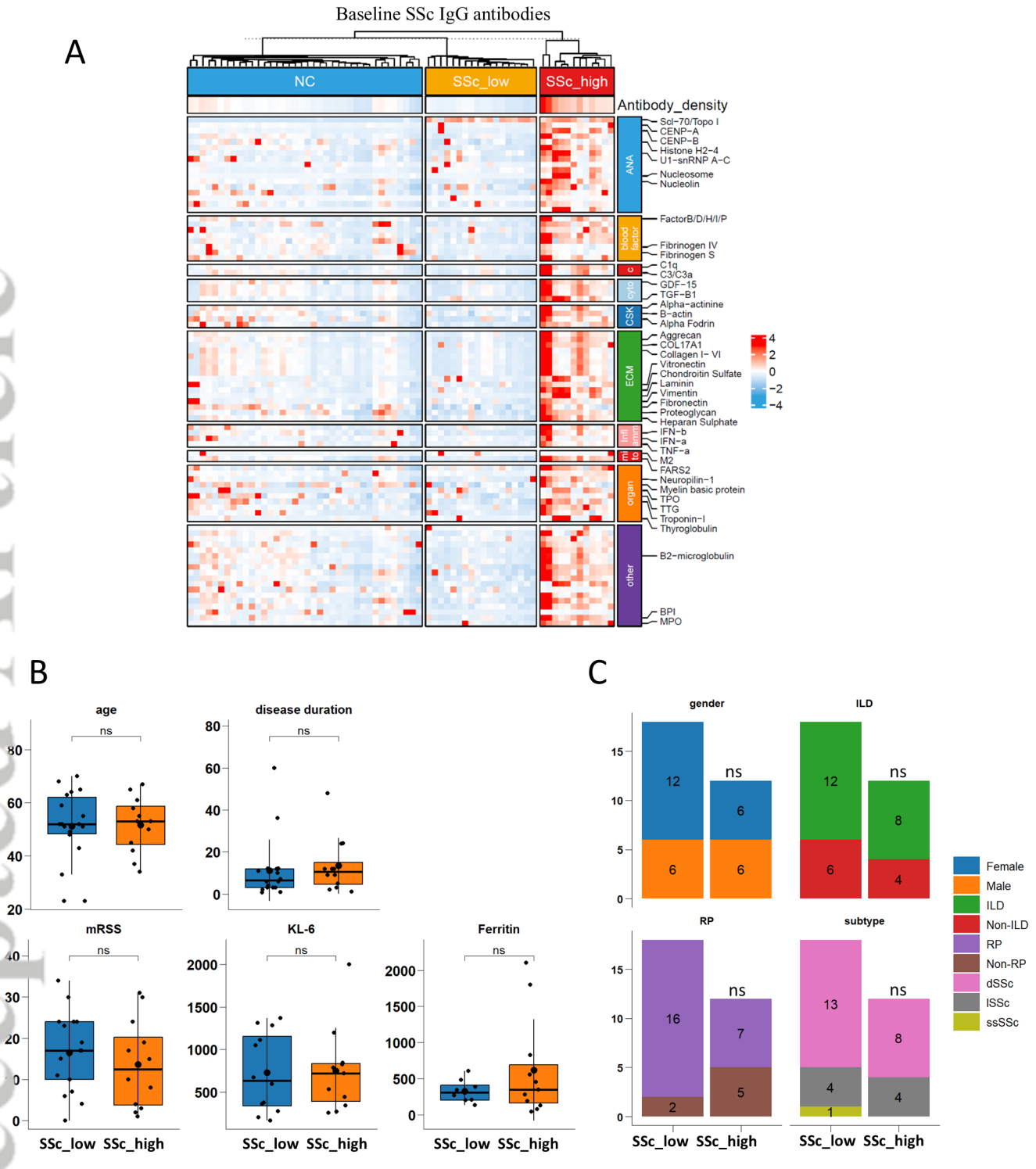

ART_43354_f1_Figure 1.TIF

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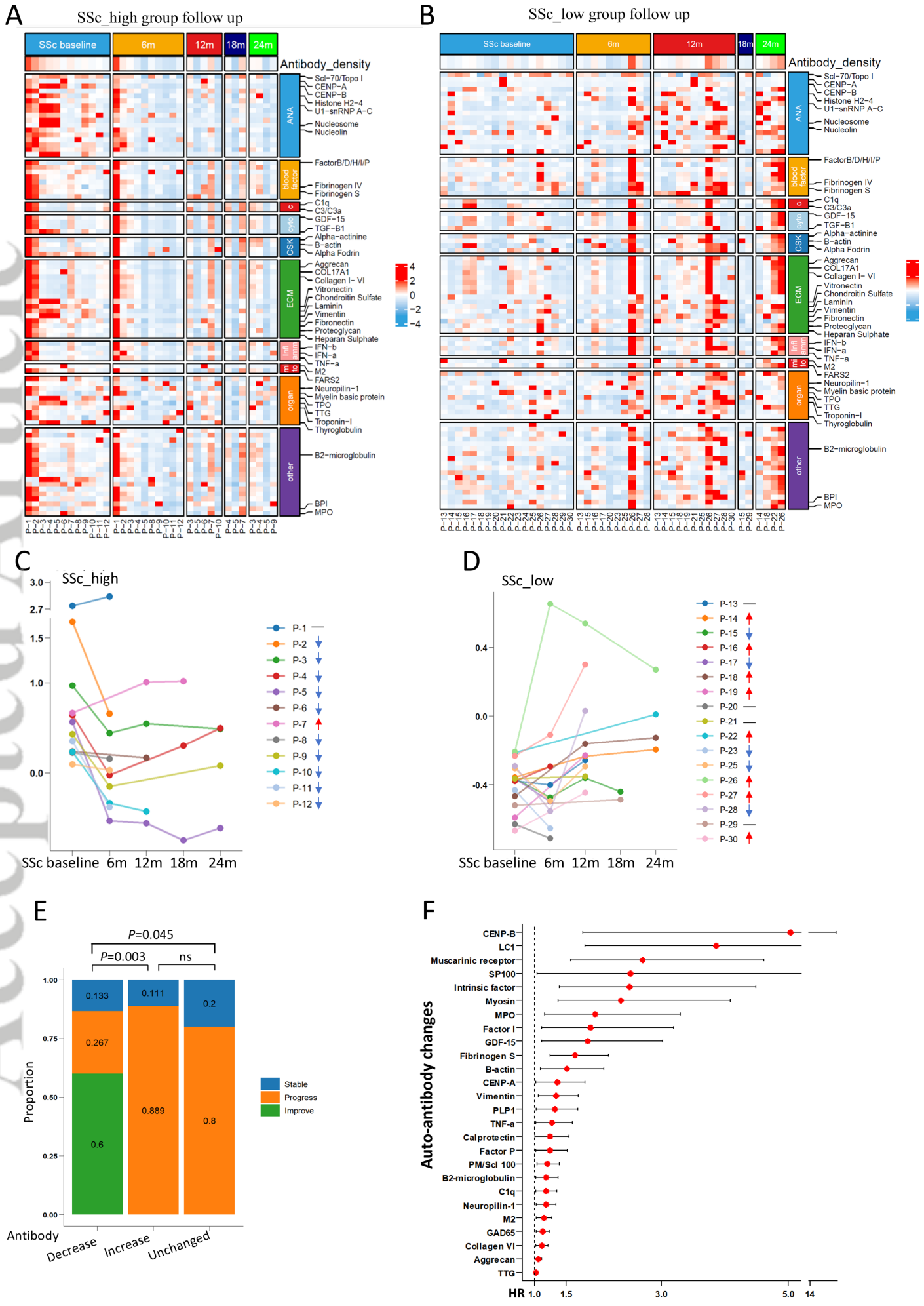

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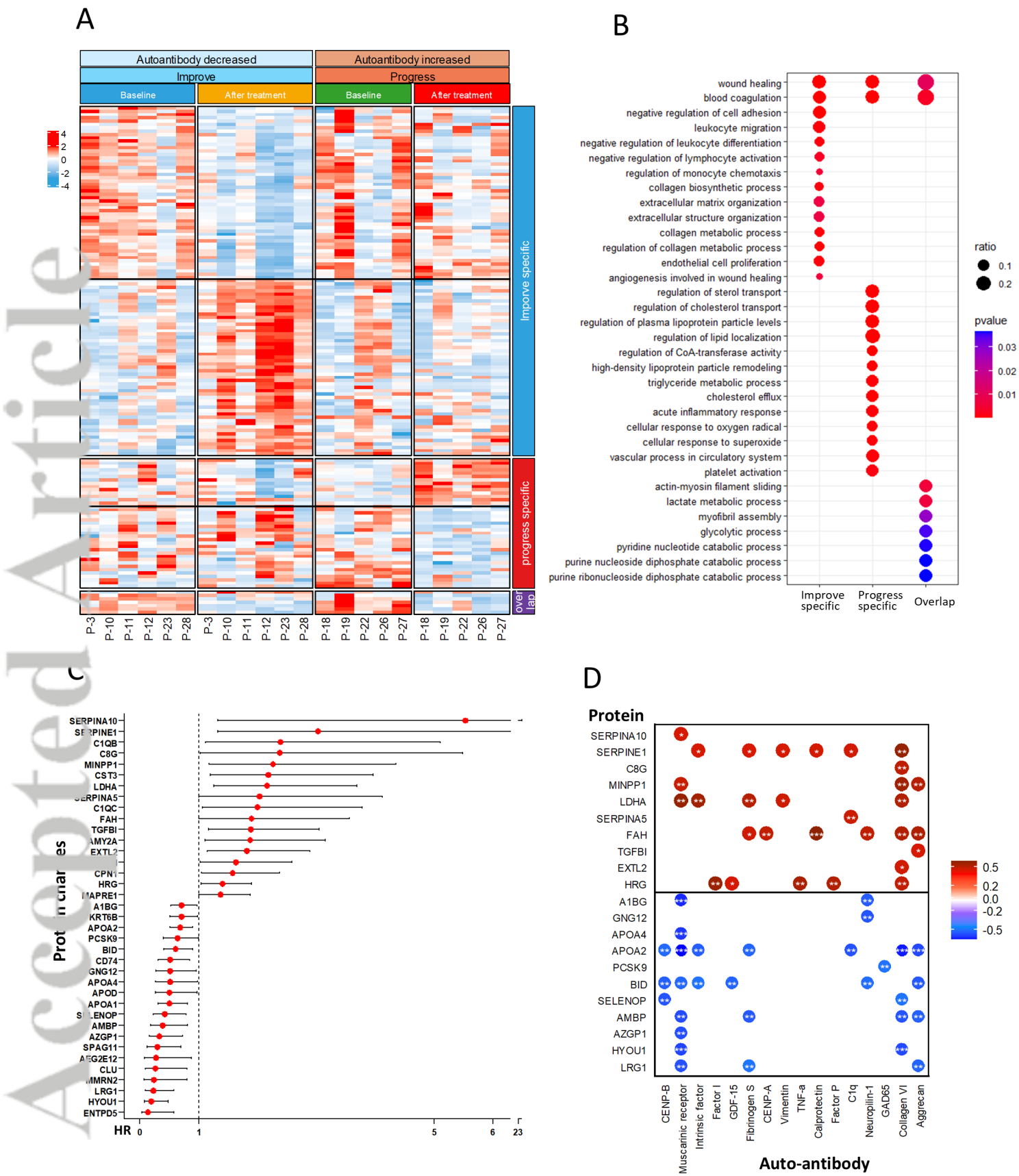

ART_43354_f3_Figure 3.TIF

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